

AWS Notes

Cloud, IAM, EC2, S3, VPC, Lambda, databases, containers, monitoring and interview revision.

Research-based study notes • Updated September 2026

1. Cloud & AWS Fundamentals

Cloud computing provides on-demand access to compute, storage, networking, databases and other resources, typically with usage-based pricing. AWS is a broad cloud platform organized into Regions and Availability Zones.

Regions, Availability Zones and Edge

A Region is a geographic AWS area. An Availability Zone (AZ) is an isolated location within a Region. Multi-AZ designs improve availability. Edge locations support services such as CloudFront for low-latency content delivery.

Exam idea: Region = geographic area; AZ = isolated infrastructure location inside a Region; edge = closer-to-user delivery point.

2. IAM — Identity and Access Management

IAM controls authentication and authorization. A principal can be a user, role, or AWS service. Policies define what actions are allowed or denied on resources.

Policy structure

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["s3:GetObject"],
    "Resource": "arn:aws:s3:::my-bucket/*"
  }]
}
```

Important concepts: identity-based policies attach to identities; resource-based policies attach to resources; roles provide temporary credentials and are preferred for applications over embedding long-lived access keys. MFA adds an additional authentication factor.

Least privilege

Grant only the permissions required for a task. Avoid broad policies such as AdministratorAccess for normal application workloads. Review permissions and use conditions, tags and resource-level ARNs where supported.

3. Amazon EC2

EC2 provides resizable virtual servers. An instance is launched from an AMI and uses an instance type defining compute, memory, networking and other capabilities.

Core terms

- AMI: template used to launch an instance.
- Instance type: hardware profile such as vCPU, memory and network capability.
- EBS: persistent block storage commonly attached to EC2.
- Security group: stateful virtual firewall controlling inbound/outbound traffic.
- Key pair: mechanism commonly used for SSH access to Linux instances.

High availability

Use multiple instances across AZs and combine them with Elastic Load Balancing and Auto Scaling when appropriate.

4. Amazon S3

S3 is object storage. Data is stored as objects inside buckets. An object has data, metadata and a key.

Key concepts

- Bucket names are globally unique within the S3 naming rules.
- Objects are addressed by keys.
- Storage classes trade cost, access frequency and retrieval characteristics.
- Versioning can preserve multiple versions of objects.
- Lifecycle rules can transition or expire objects automatically.
- Encryption can be configured at rest; access is controlled with IAM and bucket/resource policies.

Common architecture

Browser/app → CloudFront → S3 is a common pattern for static content. Application servers can upload generated files to S3 rather than storing them on local disks.

5. VPC & Networking

A VPC is a logically isolated network. It contains subnets, route tables, gateways and security controls.

Public vs private subnet

An internet-facing subnet generally has a route toward an Internet Gateway. A private subnet does not directly route inbound internet traffic through an Internet Gateway. Private workloads can use a NAT Gateway for outbound internet access when required.

Security groups vs NACLs

- Security groups are stateful and associated with network interfaces/instances.
- Network ACLs operate at the subnet boundary and are stateless, so return traffic must be explicitly allowed.

6. AWS Lambda & Serverless

Lambda runs code without requiring you to manage servers. A function is invoked by an event or request and AWS manages the execution environment.

Typical uses

- API backends with API Gateway
- S3 event processing
- Scheduled jobs
- Stream/queue consumers
- Lightweight automation

Important considerations

Design for short-lived, stateless execution. Package dependencies correctly, use environment variables for configuration, give the function a least-privilege execution role, and monitor logs/metrics. Cold starts can affect latency depending on runtime, configuration and workload.

7. Databases & Messaging

RDS

Amazon RDS is a managed relational database service supporting common relational engines. AWS handles many operational tasks such as provisioning, backups and patching, depending on configuration.

DynamoDB

DynamoDB is a managed NoSQL database designed around key-value/document access patterns. Partition-key choice strongly affects scalability and data distribution.

SQS, SNS and EventBridge

SQS is a queue for decoupling producers and consumers. SNS is a pub/sub messaging service. EventBridge routes events between producers and targets using event patterns and rules.

8. Containers

Amazon ECS is a container orchestration service; Amazon EKS is managed Kubernetes. Fargate provides serverless compute for containers, reducing the need to manage the underlying servers.

ECR

Amazon ECR is a container image registry. A typical flow is: build image → push to ECR → run with ECS/EKS/Fargate.

9. Monitoring, Security & Reliability

CloudWatch provides metrics, logs, alarms and operational visibility. CloudTrail records AWS API activity for auditing. AWS Config can track resource configuration and compliance state.

Reliability checklist

- Use multiple AZs for critical services.
- Back up stateful data and test restoration.
- Use health checks and automated replacement where possible.
- Avoid single points of failure.
- Use quotas, retries with backoff and idempotency for distributed systems.

10. AWS Architecture & Interview Revision

Three-tier web application

Route 53/DNS → CloudFront/Load Balancer → application tier (EC2/ECS/Lambda) → RDS/DynamoDB, with S3 for object/static assets and CloudWatch/CloudTrail for observability.

Quick interview questions

- What is IAM and why use roles?
- EC2 vs Lambda?
- S3 vs EBS?
- Security group vs NACL?
- RDS vs DynamoDB?
- SQS vs SNS?
- Public vs private subnet?

- Why use multiple AZs?
- What is least privilege?
- What does CloudWatch monitor?

These notes reflect current AWS documentation concepts. Always check the service's current limits, pricing, quotas and regional availability before deploying.

References

- AWS IAM documentation: <https://docs.aws.amazon.com/IAM/latest/UserGuide/> — IAM policies, roles, MFA and permissions. ■cite■turn0search11■turn0search4■
- AWS EC2 IAM/security documentation: <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/security-iam.html> ■cite■turn0search6■
- AWS Lambda IAM documentation: https://docs.aws.amazon.com/lambda/latest/dg/security_iam_service-with-iam.html ■cite■turn0search2■
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